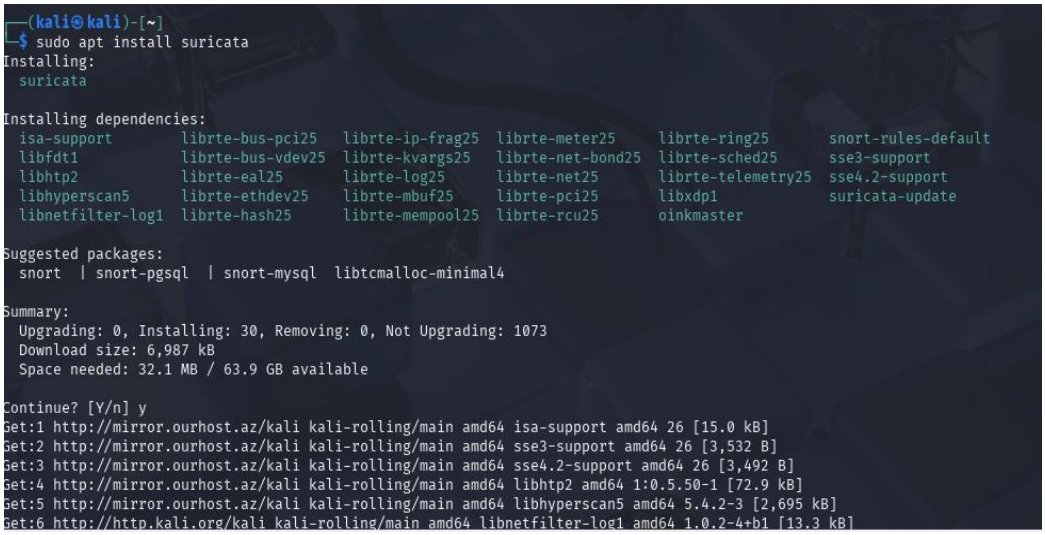
Week 4: Detecting and Analysing Network Traffic with Suricata

Prepared by Mallikah Sharp

Introduction- Suricata

Suricata is a high-performance Network IDS (Intrusion Detection System), IPS (Intrusion Prevention System), and Network Security Monitoring (NSM) tool. By following the steps below, a SOC analyst can install Suricata and create custom rules for analysing the network traffic.

1. Install Suricata and update package list. Log into the Kali Linux Virtual Machine Terminal. To update Kali run command "*sudo apt update*", then to install Surricata, run command "*sudo apt install suricata*".



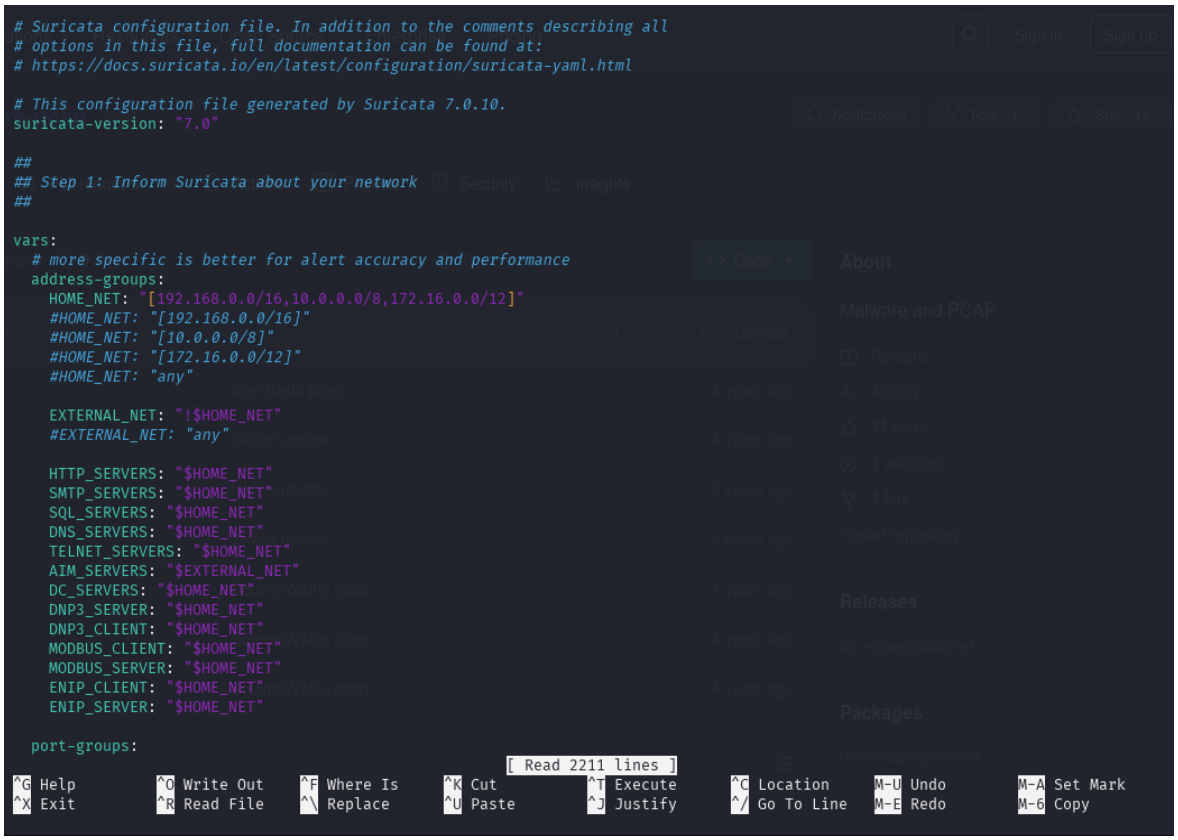
2. Create a custom rule. Run command "*sudo nano /var/lib/suricata/rules/local.rules*"

- **Add the rule to detect ICMP ping requests**: In the local.rules file, add a custom rule to detect ICMP ping requests. The rule format is based on Suricata's rule syntax.
- The following rule detects ICMP Echo Request (ping) packets:

Rule: *alert icmp any any -> any any (msg:"ICMP Ping Detected"; itype:8; sid:1000001; rev:1;)*

- alert: Action to take when the rule matches.
- icmp: Protocol to match (ICMP).
- itype:8: ICMP type 8 corresponds to Echo Request (ping).
- sid:1000001: The unique rule identifier.
- rev:1: The revision number of the rule.
- msg: "ICMP Ping Detected": Message to display when the rule is triggered.

Then update Suricata configuration. Open the file by running the command "*sudo nano /etc/suricata/suricata.yaml*".



3. Next, Press Ctrl+f to set the "*default-rule*" path. This command takes you to the files Suricata uses for rules. Now include "*local.rules*" in the rule-files section list.

```
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /var/lib/suricata/rules

rule-files:
  - suricata.rules
  - local.rules
##
## Auxiliary configuration files.
##

classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
# threshold-file: /etc/suricata/threshold.config

##
## Include other configs
##

# Includes:  Files included here will be handled as if they were in-lined
# in this configuration file. Files with relative pathnames will be
# searched for in the same directory as this configuration file. You may
# use absolute pathnames too.
```

4. Apply the changes by restarting Suricata. Run "*sudo systemctl restart suricata*".
5. Then run Suricata to verify the rules are loaded "*sudo suricata -c /etc/suricata/suricata.yaml -i eth0 -v*"

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart suricata

┌──(kali㉿kali)-[~]
└─$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0 -v
Notice: suricata: This is Suricata version 7.0.10 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
```

6. Generate ICMP traffic to trigger the rule. In a different terminal, Run "*ping -c 4 8.8.8.8*"

```
┌──(kali㉿kali)-[~]
└─$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=15.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=13.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=19.6 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=29.8 ms

── 8.8.8.8 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 13.273/19.491/29.794/6.377 ms
```

7. Verify the detection in the logs. Run "*sudo cat /var/log/suricata/eve.json | grep "ICMP Ping Detected*" (including the quotation marks)

   *** If the rule is triggered, we should see an entry with the message "ICMP Ping Detected" in the log.

***If you want to see other Suricata rules. Run the commands:

Step 1: "*cd /var/lib/suricata/rules*"

Step 2: "*ls*"

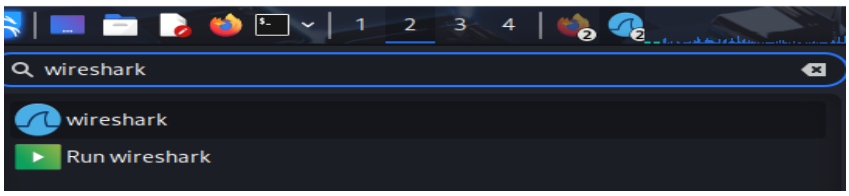
Step 3: "*cat. suricata.rules*"….*Then you will see all of the Snort rules.*

Wireshark

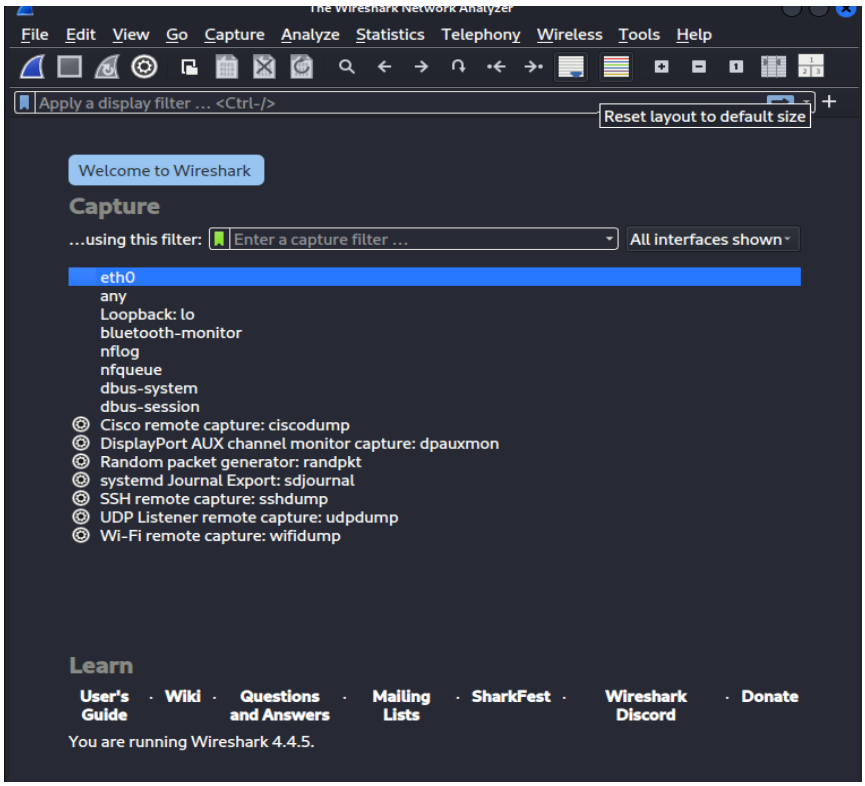
Introduction- Wireshark

Wireshark is a tool used to capture and analyse network traffic. Imagine it as a traffic camera for your network. It lets you see all the data being sent and received on your network in real-time.

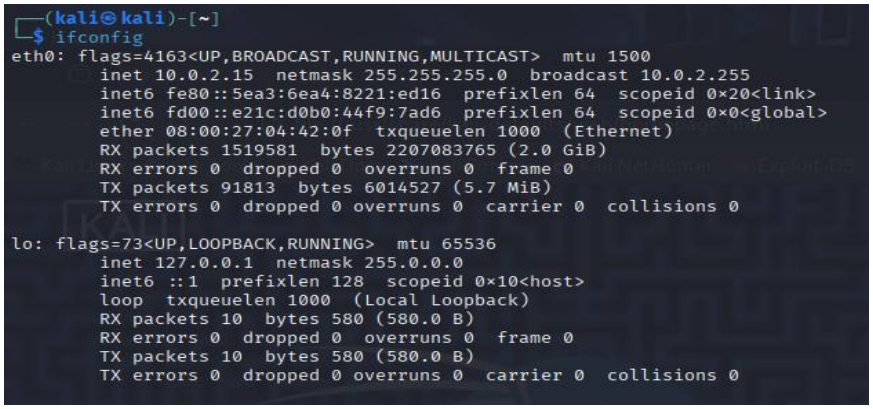
1. To begin search for Wireshark in your Kali Linux VM by typing the name in the search bar.



2. A new window will pop up as Wireshark.



3. Then in your terminal, run command: "*ifconfig*"

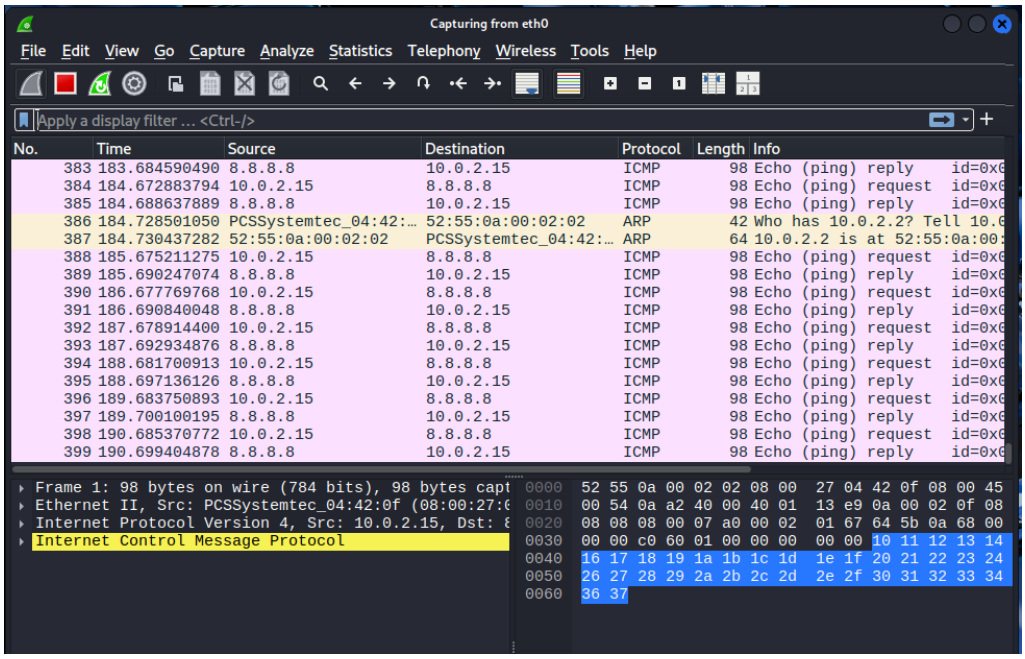


4. In the Kali terminal run command: "*ping -c 4 8.8.8.8*"
   This is the Google public DNS server.

```
┌──(kali㉿kali)-[~]
└─$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=14.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=12.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=14.6 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=12.9 ms

── 8.8.8.8 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 12.446/13.694/14.869/1.040 ms
```

5. Then in the Wireshark window, press the blue shark fin in the upper left corner. This says "Start Running Packets" This will allow us to see the ICMP (Internet Control Message Protocol). To stop the packet capture collection, press the red square next to the blue fin.



You can check the details of each packet n the left lower window pane. You can also save the collection for later as a file on your machine.

Conclusion

This project helped me to understand Suricata's role in network monitoring and detection to act as an Intrusion Detection System (IDS). I was able to write and deploy custom detection rules to trigger alerts for ICMP ping attacks. With Wireshark, I learned how to install and set up live packet captures on a network. I understand how to filter traffic in Wireshark to focus on specific types of packets and inspect packet details, such as source and destination IPS, payloads and TTL (time-to-live). This project demonstrates my hands-on experience with IDS and traffic analysis tools.